# Viva Notes: Delta Hedging of European Options

MAC F244 project (topic 12) — for every group member

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Every member should be able to answer *any* question below, not only questions about their own slides. Numbers quoted are from `results/full/tables` (seed 2026, 40,000 paths).

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## 1 The story in plain English (memorise this)

We sell a call option. The buyer pays us a premium now; at maturity we owe $\max(S_T - K, 0)$. If the stock rises a lot we lose a lot. To neutralise that risk we buy $\Delta$ shares, where $\Delta$ is how much the option's value moves per unit move in the stock. As the stock moves, $\Delta$ changes, so we must keep trading. Black–Scholes says: if you trade *continuously*, pay *no costs* and know the *true volatility*, the premium is exactly enough to pay the option off in every scenario. We relax each of those three assumptions and measure what goes wrong. Coarse trading makes errors *noisy*, at about $1/\sqrt{N}$. A wrong volatility makes errors *biased*, and more trading cannot fix that. Costs make frequent trading *expensive*, so there is a best frequency.

# 2 Symbols and units

| Symbol | Meaning | Units |
|---|---|---|
| $S_t$ | stock price at time $t$ | currency (we use $S_0 = 100$) |
| $K$ | strike | currency |
| $T,\ \tau = T - t$ | maturity, time remaining | years |
| $r$ | continuously compounded risk-free rate | per year |
| $\mu$ | real-world expected return (drift) | per year |
| $\sigma$ | volatility of log returns | per $\sqrt{\text{year}}$ |
| $\sigma_h$ | volatility the hedger *uses* | per $\sqrt{\text{year}}$ |
| $N,\ \Delta t = T/N$ | number of rebalances, interval | –, years |
| $\phi_k$ | shares held after rebalancing at $t_k$ | shares |
| $B_k$ | cash account after rebalancing at $t_k$ | currency |
| $c$ | proportional cost per side (0.001 = 10 bp) | fraction of traded value |
| $\Delta, \Gamma, \mathcal{V}$ | $\partial C/\partial S$, $\partial^2 C/\partial S^2$, $\partial C/\partial \sigma$ | –, 1/currency, currency per unit $\sigma$ |

# 3 Mathematics, step by step

## 3.1 GBM and the two measures

$dS = \mu S\,dt + \sigma S\,dW$ under $\mathbb{P}$. Apply Itô to $\ln S$: $d\ln S = (\mu - \frac{1}{2}\sigma^2)dt + \sigma dW$, so $S_{t+\Delta t} = S_t \exp\{(\mu - \frac{1}{2}\sigma^2)\Delta t + \sigma\sqrt{\Delta t} Z\}$ *exactly*, for any $\Delta t$ (no Euler error). Under the risk-neutral measure $\mathbb{Q}$ (Girsanov, with market price of risk $\lambda = (\mu - r)/\sigma$), $dS = rS\,dt + \sigma S\,dW^{\mathbb{Q}}$. **Why both?** Prices are $\mathbb{Q}$-expectations of discounted payoffs. Hedging P&L happens in the real world, so we simulate it under $\mathbb{P}$. The validation shows that the $\mathbb{Q}$-Monte-Carlo price matches BS ($z = -0.35$) and that the same estimator under $\mathbb{P}$ with $\mu = 25\%$ does not ($z = 401$).

## 3.2 Itô's lemma, replication and the PDE

For $V(t, S)$: $dV = \big(V_t + \mu S V_S + \frac{1}{2}\sigma^2 S^2 V_{SS}\big)dt + \sigma S V_S\,dW$. Hold the option and short $V_S$ shares: $\Pi = V - V_S S$, so $d\Pi = dV - V_S dS = (V_t + \frac{1}{2}\sigma^2 S^2 V_{SS})dt$. This is riskless, so no-arbitrage requires $d\Pi = r\Pi\,dt$:

$$V_t + rSV_S + \tfrac{1}{2}\sigma^2 S^2 V_{SS} - rV = 0, \quad V(T, S) = (S - K)^+.$$

$\mu$ has disappeared, and that is why the price does not depend on expected returns. The solution is the BS formula (via Feynman–Kac, $V = e^{-r\tau}\mathbb{E}^{\mathbb{Q}}[(S_T - K)^+]$).

## 3.3 Price, delta and maturity

$d_{1,2} = \frac{\ln(S/K) + (r \pm \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}$, $C = SN(d_1) - Ke^{-r\tau}N(d_2)$, $\Delta = N(d_1)$, $\Gamma = \varphi(d_1)/(S\sigma\sqrt{\tau})$, $\mathcal{V} = S\varphi(d_1)\sqrt{\tau}$. At $\tau \to 0$, $d_1 \to \pm\infty$ depending on whether $S \gtrless K$, so $\Delta \to \mathbf{1}\{S > K\}$ and $C \to (S - K)^+$. The formula divides by $\sqrt{\tau}$, so the code handles $\tau = 0$ separately and never evaluates delta at $T$. Near maturity ATM, $\Gamma \sim 1/\sqrt{\tau}$ blows up: this is where discrete hedging hurts most. Put–call parity $C - P = S - Ke^{-r\tau}$ is model-free (a forward contract), so it is an independent check.

## 3.4 Self-financing bookkeeping (derive this on the board)

Self-financing means that, between dates, no money enters or leaves except interest. At a rebalance the change in stock holding is paid for in cash:

$$\begin{aligned} t_0: &\quad B_0 = V_0 - \phi_0 S_0 - c|\phi_0| S_0 \\ t_k: &\quad B_k = B_{k-1} e^{r\Delta t} - (\phi_k - \phi_{k-1}) S_k - c|\phi_k - \phi_{k-1}| S_k \\ T: &\quad \mathrm{PnL}_T = B_{N-1} e^{r\Delta t} + \phi_{N-1} S_T (1 - c) - (S_T - K)^+. \end{aligned}$$

**Gains form** ($c = 0$): unroll the recursion. $B_{N-1}e^{r\Delta t} = V_0 e^{rT} - \sum_k (\phi_k - \phi_{k-1}) S_k e^{r(T - t_k)}$; summation by parts gives $\mathrm{PnL}_T = V_0 e^{rT} + \sum_{k=0}^{N-1} \phi_k \big(S_{k+1} - S_k e^{r\Delta t}\big) e^{r(T - t_{k+1})} - (S_T - K)^+$. Interpretation: premium grown at $r$, plus the excess return of the stock over cash on each period's holding, minus the payoff. The code computes both forms and they agree to $10^{-13}$.

**Costs are linear.** Delta ignores $c$, so the trades are identical for all $c$ and $\mathrm{PnL}_T(c) = \mathrm{PnL}_T(0) - c\,\mathrm{TV}$, where TV is the forward value of all traded notional.

### 3.5 Worked example (in `results/validation/manual_ledger_cost0.0.csv`)

$K = 100$, $T = 0.25$, $r = 5\%$, $\sigma = 20\%$, $N = 4$, path $100 \to 103 \to 98 \to 101 \to 104$. Premium 4.614997.

| $k$ | $t$ | $S$ | $\Delta$ held | trade | cash for trade | interest | cash after |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 100 | 0.569460 | +0.569460 | −56.946018 | 0 | −52.331021 |
| 1 | 0.0625 | 103 | 0.688948 | +0.119488 | −12.307226 | −0.163790 | −64.802037 |
| 2 | 0.125 | 98 | 0.435666 | −0.253281 | +24.821577 | −0.202823 | −40.183284 |
| 3 | 0.1875 | 101 | 0.612755 | +0.177089 | −17.885941 | −0.125769 | −58.194993 |
| 4 | 0.25 | 104 | 0 | −0.612755 | +63.726513 | −0.182144 | 1.349376 |

At $k = 4$ we also pay the payoff 4: $-58.194993\, e^{0.05/16} + 0.612755 \times 104 - 4 = 1.349376$. Check one line by hand: $-52.331021 \times (e^{0.003125} - 1) = -0.163790$. With $c = 10$ bp the P&L is 1.172645; the difference 0.176731 is $0.001 \times$ TV. *Why positive?* On this path the stock moved less than $\sigma = 20\%$ implies on several steps, so the short-gamma position gained. Another path could lose.

### 3.6 Why discrete hedging is not exact

Over $[t_k, t_{k+1}]$, Taylor-expand $C$: $\delta C \approx C_t \Delta t + \Delta\, \delta S + \frac{1}{2}\Gamma(\delta S)^2$. The seller is short the option and long $\Delta$ shares financed at $r$. Using the PDE to eliminate $C_t$, the one-step P&L is

$$\epsilon_k \approx \tfrac{1}{2}\Gamma_k S_k^2 \Big[\sigma^2 \Delta t - \big(\tfrac{\delta S_k}{S_k}\big)^2\Big] = \tfrac{1}{2}\Gamma_k S_k^2 \sigma^2 \Delta t\, (1 - Z_k^2).$$

$\mathbb{E}[1 - Z^2] = 0$, so there is no bias, and $\mathrm{Var}(1 - Z^2) = 2$, so $\mathrm{Var}\,\epsilon_k = \frac{1}{2}\Gamma_k^2 S_k^4 \sigma^4 \Delta t^2$. The steps are uncorrelated, so the variances add: $\mathrm{Var}\,\mathrm{PnL}_T \approx \frac{1}{2}\sigma^4 \Delta t\, \mathbb{E}\int_0^T \Gamma^2 S^4 dt = O(1/N)$. In continuous time, $\sum(\delta S/S)^2 \to \sigma^2 T$ (quadratic variation), and every $\epsilon_k$ term vanishes.

**Kamal–Derman.** For ATM, freeze $S \approx S_0$ and use $\Gamma_t \approx \Gamma_0 \sqrt{T/(T-t)}$. Then $\sum_k \frac{1}{2}(\Gamma_k S_k^2 \sigma^2 \Delta t)^2 \approx \frac{\pi}{4} N (\Gamma_0 S_0^2 \sigma^2 \Delta t)^2$, using $\int_0^T \frac{dt}{\sqrt{T^2 - t^2}} = \pi/2$. With $\mathcal{V} = S_0^2 \Gamma_0 \sigma T$ this gives sd $\approx \sqrt{\pi/4}\, \sigma \mathcal{V}/\sqrt{N}$. Base case: $\mathcal{V} = 19.644$, so at $N = 63$ the prediction is 0.4387 and the simulation gives 0.4341.

### 3.7 Wrong volatility

Price and hedge at $\sigma_h$. The same step, with $C_t$ eliminated using the PDE at $\sigma_h$, gives $\epsilon_k \approx \frac{1}{2}\Gamma_k^h S_k^2 [\sigma_h^2 \Delta t - (\delta S/S)^2]$. Its mean is now $\frac{1}{2}\Gamma^h S^2 (\sigma_h^2 - \sigma^2)\Delta t \neq 0$. Summing and growing to $T$: $\mathrm{PnL}_T \approx \frac{1}{2}\int_0^T e^{r(T-t)} \Gamma_h S^2 (\sigma_h^2 - \sigma^2)\, dt$. The sign depends only on $\sigma_h^2 - \sigma^2$ because $\Gamma > 0$ for a call. The *size* is path-dependent (how long the path spends near the strike). That is why the sd does not go to 0 when $N \to \infty$.

### 3.8 Transaction costs and Leland

The expected traded shares per step are $\approx \Gamma S \cdot \mathbb{E}|\delta S| = \Gamma S^2 \sigma \sqrt{\Delta t}\sqrt{2/\pi}$, so the cost per step is $\approx c\, \Gamma S^2 \sigma \sqrt{2/\pi}\sqrt{\Delta t}$ and the total cost is $\propto c\sqrt{N}$. Leland writes this as an extra $\frac{1}{2}\Gamma S^2 \cdot \sigma^2 \sqrt{2/\pi}\, k/(\sigma\sqrt{\Delta t})\, \Delta t$ with round-trip $k = 2c$, which is the same as using $\sigma_L^2 = \sigma^2(1 + \sqrt{2/\pi}\, k/(\sigma\sqrt{\Delta t}))$. Our simulation shows the remaining mean loss under Leland (0.12 at 10 bp) equals the initial-purchase plus liquidation cost (0.121). The model does not cover those trades.

## 4 Code map (equation → function)

| | |
|---|---|
| Exact GBM step | `gbm.simulate_gbm` |
| $C, P, \Delta, \Gamma, \mathcal{V}$, $\tau = 0$ cases | `black_scholes.py` |
| Bookkeeping $B_0, B_k, \mathrm{PnL}_T$; TV; costs | `hedge.hedge_short_call` |
| Gains form (independent check) | `hedge.pnl_gains_form` |
| Hand ledger | `hedge.hedge_ledger` |
| Gamma formula for wrong $\sigma$ | `hedge.predicted_pnl_misspecified` |
| Leland $\sigma_L$ | `black_scholes.leland_sigma` |
| Statistics and SEs | `stats.summarize` |
| Experiments A–D / E | `experiments.py` / `realdata.py` |

**Design choices you may be asked to justify.** *Fine grid + subsampling*: every frequency uses the same Brownian path, so comparisons are paired (common random numbers). *Batches of 5,000*: limit memory, and they also serve as 8 independent replications (slope SE). *SeedSequence.spawn*: independent streams per experiment from one master seed. *Delta-method SE for the sd*: $\mathrm{SE}(s) \approx \sqrt{(m_4 - s^4)/n}/(2s)$ uses the sample fourth moment, so it remains valid for fat tails. *Bootstrap* for quantiles and ES.

# 5 Key results to quote

- A: sd 1.856 ($N = 3$) $\to$ 0.434 ($N = 63$, daily) $\to$ 0.098 ($N = 1260$). Slope $-0.488 \pm 0.001$. Kamal–Derman within 1.4% for $N \geq 63$. Daily sd is 9.4% of the premium 4.615.
- A, drift: sd 0.439/0.434/0.418 for $\mu = 5/10/25\%$ at $N = 63$. The effect is small but detectable (discrete-time only).
- B: means $-1.94, -0.98, 0, +0.99, +1.98$ for $\sigma_h = 10..30\%$ (theory identical to 2 dp). The bias does not shrink with $N$. If the premium is charged at the true $\sigma$, the mean is about 0.
- C: RMSE-optimal $N = 1260/252/126/63/21$ for 0/5/10/20/50 bp. Leland mean $-0.12$ at 10 bp for all $N$, equal to the setup and unwind cost.
- D: ATM sd 0.32 vs 0.98 (15% vs 45% vol). OTM ($K = 120$) sd/premium 254% vs 30%. Absolute sd flat in maturity.
- E: IBM 29%, MSFT 34%, GOOG 39%, AAPL 55%, AMZN 59% (2000–10 monthly); AAPL 24% (2015–17 daily). Backtest corr(P&L, variance gap) = 0.87.

# 6 Question bank with model answers

**Q1. Why does the option price not depend on $\mu$?** The hedged portfolio is riskless, so it must earn $r$. $\mu$ cancels in $d\Pi$. Risk preferences are absorbed by replication.

**Q2. Then why simulate under $\mathbb{P}$?** The P&L is realised along real-world paths. In continuous time the result is $\mu$-independent. In discrete time $\mu$ has a small effect (sd falls 4.6% from $\mu = 5\%$ to 25%) because paths drift away from the strike, where gamma is lower.

**Q3. Is the exact GBM step an approximation?** No. $\ln S$ increments are exactly normal, so any step size is exact in distribution. Validation: $E[S_T]$ and $\mathrm{Var} \ln S_T$ match theory with 1 step and with 252 steps.

**Q4. What does self-financing mean in your code?** The only cash flows are the premium at 0, trades, costs, interest and the final payoff. Line `cash -= trade * S_k` is exactly that.

**Q5. Why is the P&L at $T$ and not at 0?** So that all cash flows are in the same time-$T$ money. Discounting by $e^{-rT}$ (a factor of 0.988) changes nothing qualitatively.

**Q6. Sign convention?** Positive means the seller gains.

**Q7. Why does the mean error vanish but the sd does not?** Each step's error is $\propto (1 - Z^2)$, which has mean 0 and variance 2.

**Q8. Why $1/\sqrt{N}$?** There are $N$ independent errors, each of size $\propto \Delta t = T/N$, so the variance is $N \cdot (T/N)^2 \propto 1/N$.

**Q9. Your slope is $-0.488$, not $-0.5$. Is your code wrong?** No. $-\frac{1}{2}$ is the asymptotic slope. Coarse grids ($N = 3, 9$) are outside the asymptotic regime: the Kamal–Derman approximation overstates the sd by 8% at $N = 3$. The slope fitted on $N \geq 63$ alone is $-0.497$.

**Q10. Is every error statistic monotone in $N$?** Not necessarily, and |mean| is not in our sample. We report monotonicity as an observation, not a theorem.

**Q11. Why are the tails fat?** The error is a sum of $\Gamma$-weighted $(1 - Z^2)$ terms, and the weights are random and large near the strike close to expiry. Paths that end near $K$ dominate the losses.

**Q12. Why is $N = 3$ slightly biased?** The second-order Taylor expansion fails for large moves, and higher-order terms do not average out. The bias is small ($-0.022 \pm 0.009$).

**Q13. Why is over-estimating $\sigma$ good for the seller?** The seller collects a larger premium, and the gamma formula shows a positive expected P&L whenever $\sigma_h > \sigma$ (El Karoui et al. 1998: a super-hedge if the hedge vol dominates).

**Q14. If the premium is fair but the delta is wrong?** The mean is about 0, but the sd is unchanged and does not vanish as $N \to \infty$. The wrong delta adds risk, not expected loss.

**Q15. Why do costs grow like $\sqrt{N}$?** The shares traded per step are $\propto \Gamma |\delta S| \propto \sqrt{\Delta t}$, and there are $N$ steps, so the total is $\propto N\sqrt{T/N} = \sqrt{NT}$.

**Q16. Why is your Leland mean not zero?** Leland's correction prices only *rebalancing* costs. We also charge the initial purchase and the final liquidation (0.121 at 10 bp), which explains the residual exactly.

**Q17. Why $k = 2c$?** Leland defines $k$ as the round-trip cost. We charge $c$ per side.

**Q18. Is RMSE the right objective?** It is one convention that combines bias and spread. A risk-averse seller might minimise mean + $\lambda$·sd, or ES. The optimum would move.

**Q19. How do you know your Monte Carlo is accurate enough?** Every statistic carries an SE. With 40,000 paths the sd at $N = 63$ is $0.434 \pm 0.002$. Two independent seeds agree within 3 SE.

**Q20. Why do OTM options have huge relative errors?** The premium is tiny (0.03), but if the stock reaches the strike, gamma becomes large. The absolute error is small, but relative to the premium it is large.

**Q21. Why is the absolute error flat in maturity at fixed daily rebalancing?** sd $\approx \sqrt{\pi/4}\,\sigma\mathcal{V}/\sqrt{N}$, with $\mathcal{V} \propto \sqrt{T}$ (ATM) and $N \propto T$, so $T$ cancels.

**Q22. Is your real data 'market data'?** It is secondary-source historical prices (vega-datasets and plotly/datasets mirrors). We used no option quotes. The premiums are model prices. The results are illustrations, not trading evidence.

**Q23. Why are overlapping windows a problem?** Consecutive windows share 62 of 63 days, so the 380 windows are far from independent. Only 7 are non-overlapping.

**Q24. Biggest limitation?** Constant volatility. Real volatility is stochastic and jumps exist, and then delta hedging alone cannot replicate the option.

**Q25. What would you do next?** Stochastic volatility (Heston); hedging with the Whalley–Wilmott band instead of fixed-time rebalancing; using real option quotes; adding gamma hedging with a second option.

# 7 Honest limits of what the plots show

Fig. 1 shows the error distribution only in the BS world. Fig. 2b shows agreement with a continuous-time approximation, not proof of it. Fig. 3's optimum depends on choosing RMSE. Fig. 4 uses two stylised regimes, not calibrated assets. Fig. 5 is one stock, two years, model premiums, and overlapping windows.